

Lab 1: Introduction to R, RStudio, and Quarto

Math 141

Welcome to our first lab assignment! In this lab, you will practice:

1. Warming up with fundamental commands in R.
2. Running R code from chunks.
3. Viewing and describing data frames in R.
4. Rendering a Quarto document.

To-Do:

- Integrate the previous warm up file I had with this, start from very basics.
- Make a problem about saving!!!
- Make a problem about rendering and finding the PDF and downloading it (if on server)
- Make Problem 1a a “hello world” and have them practice the different ways of running the line of code.
- Notes for which parts I want to do together on the screen, any timing for working together then showing results

Problems

Problem 1

In this problem, we will explore fundamental commands and behavior in R.

- a. In the R code chunk below, create a variable called `y` with the value 7. In R, we use `<-` to assign values to variable names, like `x <- 1`. *Pro tip: To run a single line of code, put the cursor on the same line as the code and hit Command+Enter (for a Mac) or Control+Enter (for a PC). To run all the code in a chunk, put the cursor on a line of code and hit Command+Shift+Enter (for a Mac) or Ctrl+Shift+Enter (for a PC).*

- b. Below, create a new code chunk. Create another variable called `x` with the value of 5, and a third variable called `z` with the value of `x` multiplied by `y`.
- c. Run the chunk below. Where does the answer appear? Find the console pane (usually the bottom left of the RStudio window), and run the same command, `2*50`, in the console. Where does the answer appear now?

```
2*50
```

- d. Create another new chunk below, and use the `sum()` function to add `x`, `y`, and `z`. If you aren't sure how to use `sum()`, try running `?sum` in the console to pull up the help page.
- e. Find the environment pane in your RStudio window (usually in the upper right). What do you see there?
- f. In the environment pane, click the little broom icon in the upper bar. Click yes on the window that opens. What happened to the environment? What happens if you now try to run your chunks from steps (b) or (d)?
- g. Find the "Run" button at the upper right corner of this chunk, and select "Run all chunks above." What happened to the environment?
- h. Let's practice using exponents to create another variable, `s`. Run the chunk below one line at a time using Command+Enter (for a Mac) or Control+Enter (for a PC), and type out the value associated with each line. Why does each line affect the environment differently? Where did you find the values?

```
s <- 7^7  
7^5
```

- i. Finally, let's practice making a vector and creating character data types. In R, we create vectors using the function `c()`. For example, try running `c(1, 2, 3)` in the console. All elements of a vector must have the same data type (ex. all numeric, or all character). We create characters by writing `"some text"`. In a new chunk below, create a vector called `a` with three character elements, with each element representing one of your favorite genres of music. What information about this object is stored in the environment?

Problem 2

In this problem, we are going to explore an existing data frame in R.

- a. For data frames in general, what does a row represent? What does a column represent?

- b. Run the following code to load the `openintro` library/package and to grab the `yrbss` dataset.

```
#load the openintro package and the yrbss dataset
library(openintro)
library(dplyr)
data(yrbss)
```

- c. Run the following, and begin exploring the data called `yrbss`. (Leave in the `eval: false`.) Some useful functions to get started include: `head()` and `str()`. You don't need to write any written response for this part, just use this space to explore the data set and answer the following questions.

Pro tip: For datasets in packages, we can also run `?` followed by the dataset name to open up the documentation file in the Help pane that describes the dataset and provides information on the variables.

```
View(yrbss)
?yrbss
```

- d. What does a row of the data frame represent? How many rows are in the data frame?
- e. What does a column of the data frame represent? How many columns are in the data frame?
- f. Is `age` recorded as a categorical or a quantitative variable? How about `hours_tv_per_school_day`? Explain your reasoning.
- g. We will learn to be careful with how we treat missing values, which are typically labeled as `NA` in the data frame. What are two possible reasons why the variable `text_while_driving_30d` might be recorded as `NA`?